Quiz-1

CSD205: Discrete mathematics **Time: 55 Min**

Each question carries one marks and question 12 caries 2 marks

1: Consider two well-formed formulas in prepositional logic

F1:$P \Rightarrow \sim P$ and F2:$(P \Rightarrow \sim P) \vee (\sim P \Rightarrow P)$

Which of the following statements is correct?

A). F1 is satisfiable, F2 is valid B). F1 unsatisfiable, F2 is satisfiable

C). F1 is unsatisfiable, F2 is valid D). F1 and F2 are both satisfiable

2. Which of the following propositions is a tautology?

A). $((p \vee q) \Rightarrow p)$ B). $(p \vee (q \Rightarrow p))$ C). $(p \vee (p \Rightarrow q))$ D). $(p \Rightarrow (p \Rightarrow q))$

3. Consider the following formula and its two interpretations I_1 and I_2

$\alpha : (\forall x)[P_x \Leftrightarrow (\forall y)[Q_{xy} \Leftrightarrow \neg Q_{yy}]] \Rightarrow (\forall x)[\neg P_x]$

I_1: 'Domain: the set of natural numbers'

P_x: 'x is a prime number'

Q_{xy}: 'y divides x'

I_2: Same as I_1 except that P_x: 'x is a composite number'.

Which of the following statements is true?

A). I_1 satisfies α, I_2 does not B). I_2 Satisfies α, I_1 does not

C).Neither I_2 nor I_1 satisfies α D). Both I_1 and I_2 satisfy α

4. Identify the correct translation into logical notation of the following assertion.

Some boys in the class are taller than all the girls

Note: $taller(x, y)$ is true if x is taller than y.

A). $(\exists x)[Boy(x) \rightarrow (\forall y)(Girl(y) \wedge Taller(x,y))]$

B). $(\exists x)[Boy(x) \wedge (\forall y)(Girl(y) \wedge Taller(x,y))]$

C). $(\exists x)[Boy(x) \rightarrow (\forall y)(Girl(y) \rightarrow Taller(x,y))]$

D). $(\exists x)[Boy(x) \wedge (\forall y)(Girl(y) \rightarrow Taller(x,y))]$

5. Let $a(x,y), b(x,y)$ and $c(x,y)$ be three statements with variables x and y chosen from some universe. Consider the following statement:

$(\exists x)(\forall y)[(a(x,y) \wedge b(x,y)) \wedge \neg c(x,y)]$

Which one of the following is its equivalent?

A). $(\forall x)(\exists y)[(a(x,y) \vee b(x,y)) \Rightarrow c(x,y)]$

B). $(\exists x)(\forall y)[(a(x,y) \vee b(x,y)) \wedge \neg c(x,y)]$

C). $\neg[(\forall x)(\exists y)[(a(x,y) \wedge b(x,y)) \Rightarrow c(x,y)]]$

D). $\neg[(\forall x)(\exists y)[(a(x,y) \vee b(x,y)) \Rightarrow c(x,y)]]$

6. What is the first order predicate calculus statement equivalent to the following?

Every teacher is liked by some student

A). $\forall x[teacher(x) \rightarrow (\exists y)(student(y) \rightarrow likes(y,x))]$

B). $\forall x[teacher(x) \rightarrow (\exists y)(student(y) \wedge likes(y,x))]$

C). $\exists y \forall x[teacher(x) \rightarrow (student(y) \wedge likes(y,x))]$

D). $\forall x[teacher(x) \wedge (\exists y)(student(y) \rightarrow likes(y,x))]$

7. Suppose the predicate $F(x,y,t)$ is used to represent the statement that person x can fool person y at time t. which one of the statements below expresses best the meaning of the formula " $\forall x \exists y \exists t(\neg F(x,y,t))$ "?

A).Everyone can fool some person at some time.

B). No one can fool everyone all the time.

C).Everyone cannot fool some person all the time.

D). No one can fool some person at some time.

8: Solve the recurrence relation

$$T(n) = T(n-1) + n;\ and\ T(1) = 1$$

9: Solve the recurrence relation

$$T(n) = T\left(\frac{n}{2}\right) + 1;\ \text{and}\ T(1) = 1$$

10: Solve the following recurrence Relation:

$$x_n = 2x_{n-1} - 1;\ where\ n > 1;\ and\ x_1 = 2;$$

11: The solution of the recurrence relation is

$$T(2^k) = 3T(2^{k-1}) + 1;\ and\ T(1) = 1;$$

A). 2^k B). $\left(\frac{3^{k+1} - 1}{2}\right)$ C). $3^{\log_2 k}$ D). $2^{\log_3 k}$

12: The Island has two types of inhabitants, “knights” who always tell the truth, and “knaves” who always lie.

It is rumoured that there is gold buried on the island. You ask one of the natives, A, whether there is gold on the island. He makes the following response: “There is gold on this island if and only if I am a knight.”

The problem is

(a) Can it be determined whether A is a knight or a knave?

(b) Can it be determined whether there is gold on the island?

13: What is time complexity of fun ()?

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int fun(int n)
{
  int count = 0;
  for (int i = n; i > 0; i /= 2)
```

```
    for (int j = 0; j < i; j++)
        count += 1;
  return count;
}
```

(A) O(n^2)
(B) O(nLogn)
(C) O(n)
(D) O(nLognLogn)

14: Explain Big-oh , theta and omega notation for complexity analysis